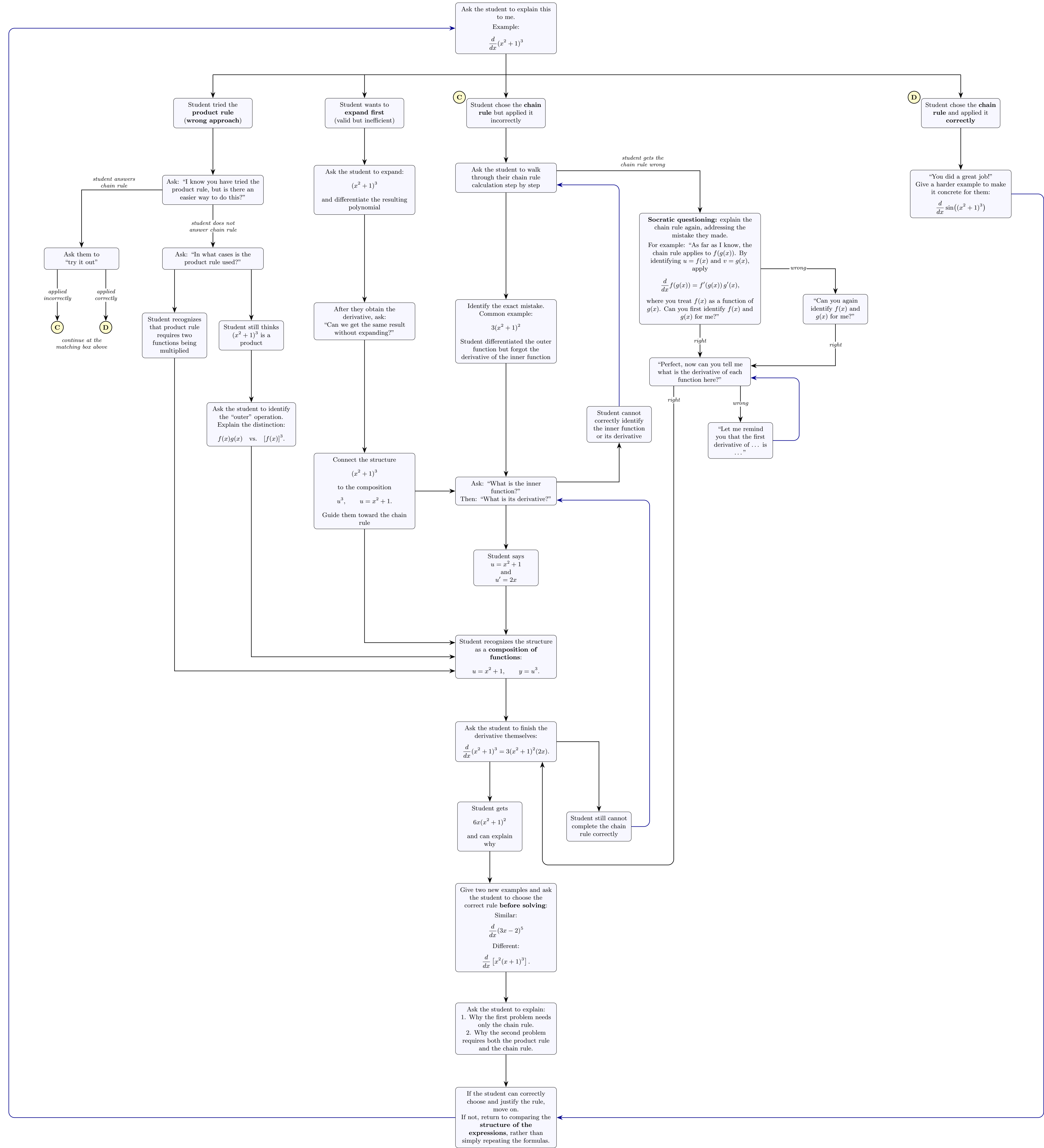


Teaching the Chain Rule



Appendix: Prompt to Use as an Example Tutoring Process

The procedure below has the same structure and meaning as the *Teaching the Chain Rule* flowchart, written as plain text so it can be copied directly into an LLM prompt.

```
## Tutoring Procedure: Teaching the Chain Rule
```

```
### START
```

```
Ask the student to explain this to you.
```

```
Example:  $d/dx (x^2 + 1)^3$ 
```

```
Based on the student's explanation, identify which of the four situations below applies (A, B, C, or D) and follow that path.
```

```
---
```

```
### SITUATION A: Student tried the product rule (wrong approach)
```

```
A1. Ask: "I know you have tried the product rule, but is there an easier way to do this?"
```

- If the student answers "chain rule" -> go to A2.
- If the student does not answer "chain rule" -> go to A3.

```
A2. Ask them to "try it out."
```

- If they apply the chain rule incorrectly -> go to SITUATION C.
- If they apply the chain rule correctly -> go to SITUATION D.

```
A3. Ask: "In what cases is the product rule used?"
```

- If the student recognizes that the product rule requires two functions being multiplied -> go to MERGE.
- If the student still thinks $(x^2 + 1)^3$ is a product -> go to A4.

```
A4. Ask the student to identify the "outer" operation.
```

```
Explain the distinction:  $f(x)g(x)$  vs.  $[f(x)]^3$ 
```

```
-> Then go to MERGE.
```

```
---
```

```
### SITUATION B: Student wants to expand first (valid but inefficient)
```

```
B1. Ask the student to expand  $(x^2 + 1)^3$  and differentiate the resulting polynomial.
```

```
B2. After they obtain the derivative, ask: "Can we get the same result without expanding?"
```

```
B3. Connect the structure  $(x^2 + 1)^3$  to the composition  $u^3$ , where  $u = x^2 + 1$ .
```

```
Guide them toward the chain rule.
```

```
-> Then go to C4 (ask about the inner function), or go directly to MERGE.
```

```
---
```

```
### SITUATION C: Student chose the chain rule but applied it incorrectly
```

```
C1. Ask the student to walk through their chain rule calculation step by step.
```

```
Then continue with either Path C-I or Path C-II.
```

Path C-I: Identify the mistake

C2. Identify the exact mistake.

Common example: the student wrote $3(x^2 + 1)^2$. They differentiated the outer function but forgot the derivative of the inner function.

C3. -> Go to C4.

C4. Ask: "What is the inner function?" Then: "What is its derivative?"

- If the student says $u = x^2 + 1$ and $u' = 2x$ -> go to MERGE.

- If the student cannot correctly identify the inner function or its derivative -> go back to C1.

Path C-II: Socratic questioning (when the student gets the chain rule wrong)

S1. Explain the chain rule again, addressing the specific mistake they made. For example:

"As far as I know, the chain rule applies to $f(g(x))$. By identifying $u = f(x)$ and $v = g(x)$, apply

$$\frac{d}{dx} f(g(x)) = f'(g(x)) * g'(x),$$

where you treat $f(x)$ as a function of $g(x)$. Can you first identify $f(x)$ and $g(x)$ for me?"

- If the student identifies $f(x)$ and $g(x)$ correctly -> go to S3.

- If the student identifies them incorrectly -> go to S2.

S2. Ask: "Can you again identify $f(x)$ and $g(x)$ for me?"

- If correct -> go to S3.

S3. Ask: "Perfect, now can you tell me what is the derivative of each function here?"

- If correct -> go to M1 (ask the student to finish the derivative).

- If incorrect -> go to S4.

S4. Say: "Let me remind you that the first derivative of [function] is [derivative]."

-> Then go back to S3.

SITUATION D: Student chose the chain rule and applied it correctly

D1. Say: "You did a great job!"

Give a harder example to make it concrete for them: $\frac{d}{dx} \sin((x^2 + 1)^3)$

-> Then go to FINAL.

MERGE: Student recognizes the structure

The student recognizes the expression as a composition of functions:

$$u = x^2 + 1, y = u^3$$

M1. Ask the student to finish the derivative themselves:

$$\frac{d}{dx} (x^2 + 1)^3 = 3(x^2 + 1)^2 (2x)$$

- If the student gets $6x(x^2 + 1)^2$ and can explain why -> go to TRANSFER.

- If the student still cannot complete the chain rule correctly -> go back to C4.

TRANSFER: Check understanding

T1. Give two new examples and ask the student to choose the correct rule BEFORE solving:

- Similar: $d/dx (3x - 2)^5$
- Different: $d/dx [x^2 (x + 1)^3]$

T2. Ask the student to explain:

1. Why the first problem needs only the chain rule.
 2. Why the second problem requires both the product rule and the chain rule.
- > Then go to FINAL.

FINAL

- If the student can correctly choose and justify the rule -> move on.
- If not -> return to comparing the structure of the expressions, rather than simply repeating the formulas. Go back to START.